

Ayush Kumar

 Ayush Kumar |  ayush-244 |  Ayush Kumar |  +919162573098  ayushku244@gmail.com

WORK EXPERIENCE

Backend Developer Intern – Seqqto Software Pvt. Ltd.

Nov 2025 – Feb 2026

- Developed **5+ production REST APIs** using Node.js and Express.js, including data ingestion endpoints supporting **downstream ML model serving pipelines** across 3 client-facing applications.
- Designed **MongoDB schemas with compound indexing and aggregation pipelines**, optimizing structured data storage for **feature extraction and analytics workflows**.
- Integrated **JWT-based authentication and third-party APIs** within Agile sprints, delivering secure backend infrastructure for **AI-backed applications**.

Summer Research Intern – SRM University-AP

June – July 2025

- Analyzed **50,000+ financial records**; applied **feature engineering** (correlation filtering, PCA, binning) and EDA to identify key credit score drivers for downstream predictive modeling.
- Built an **end-to-end ML pipeline** evaluating Logistic Regression, Random Forest, and XGBoost with **cross-validation and hyperparameter tuning**, optimizing F1-score via systematic feature selection.

PROJECTS

HearMe AI — Multi-Task NLP & RAG Platform

[Link](#)

Python, FastAPI, PyTorch, LangChain, Groq LLM, RAG, Qdrant, SentenceTransformers, HuggingFace Transformers, Next.js

- Designed a **multi-task NLP inference system** with 15+ specialized models — **RoBERTa-base** for sentiment/toxicity/emotion, **DistilBERT** for intent classification — with adaptive **LLM routing across 4 response modes** and 24+ prompt injection guardrails.
- Implemented **production RAG pipeline**: hybrid BM25 + dense vector retrieval (**BAAI/bge-base-en-v1.5, 768-dim**), **SHA256 embedding cache** reducing re-inference overhead, and multi-format document parsing (PDF/DOCX/TXT/Markdown) with 3 auto-selected chunking strategies.
- Engineered a **4-type long-term memory system** (Semantic, Episodic, Preference, Working) with cosine similarity deduplication and **importance decay scoring** for persistent cross-session personalization.

AuthenticAI — AI Image Detection with XAI

[Link](#)

Python, TensorFlow, Flask, Grad-CAM, Google Gemini API, React, Vite, Docker, GitHub Actions

- Trained a **custom CNN on AI-generated image datasets** (DALL-E, Midjourney) achieving **95%+ classification accuracy**; applied batch normalization, dropout regularization, and mixed precision to achieve **sub-100ms CPU inference**.
- Applied **Grad-CAM gradient attribution** for layer-wise heatmap visualization of discriminative image regions; integrated **Google Gemini API** to generate natural-language **XAI explanations** for non-technical stakeholders.
- Served model via **Docker-containerized Flask API** with **rate limiting (20 req/min)**, API key auth, and **GitHub Actions CI** pipeline with ESLint and Vitest automated testing.

MediBridge AI — Healthcare AI Copilot (RAG + Severity Prediction)

[Link](#)

React 18, TypeScript, Node.js, FastAPI, ChromaDB, DeepSeek LLM (OpenRouter), Socket.IO, MongoDB, Docker

- Built an **AI Clinical Copilot** using **ChromaDB vector store** and **DeepSeek LLM (OpenRouter)** for **RAG-based medical document Q&A**, ICD-10 severity classification, and intelligent **hospital-specialist matching**.
- Architected a **3-service ML inference system** (React + Node.js + FastAPI) with **Docker Compose** orchestration, isolating the AI inference layer for independent scaling and zero-downtime model updates.
- Implemented **real-time ML prediction streaming** via Socket.IO event bus across a **4-role RBAC system** with full audit logging and bed reservation lifecycle management.

EDUCATION

B.Tech (CSE) at **SRM University AP** (CGPA: 8.85/10.0)

2023 – present

Class 12th CBSE Board [Cambridge Senior Sec. School, Buxar (Bihar)]

2022

SKILLS

AI & Deep Learning	PyTorch, TensorFlow, CNN, Transformers, ViT, BERT/RoBERTa, Transfer Learning, HuggingFace, NLP, Computer Vision, LangChain
GenAI & LLMops	LLMs, RAG, LangChain, Prompt Engineering, Qdrant, ChromaDB, SentenceTransformers, BAAI/bge, Vector Search
ML Engineering	Scikit-learn, XGBoost, Random Forest, Feature Engineering, Cross-Validation, Grad-CAM, XAI, Model Evaluation and Hyperparameter Tuning
Data Science	NumPy, Pandas, EDA, PCA, Statistical Modeling, Hypothesis Testing, A/B Testing, Data Visualization
MLOps & Backend Programming	FastAPI, Flask, Docker, GitHub Actions CI/CD, REST APIs, Model Serving, Node.js, MongoDB
	Python, SQL, TypeScript, JavaScript, C++, Java

CERTIFICATIONS

- SAP Certified – SAP Generative AI Developer – SAP (2026)**
- Oracle Cloud Infrastructure 2025 Certified Data Science Professional – Oracle (2025)**
- MongoDB Certified Associate Developer (Node.js) – MongoDB (2026)**

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ACHIEVEMENTS

- Awarded a **90% merit-based tuition scholarship** at SRM University-AP for academic excellence.
- Selected as **Tech Team Member** of the **Google Developer Groups (GDG)** at SRM University-AP.